
Coastal Resource Management and Social Differences in Philippine Fishing Communities

Author(s): James F. Eder

Source: *Human Ecology*, Vol. 33, No. 2 (Apr., 2005), pp. 147-169

Published by: [Springer](#)

Stable URL: <http://www.jstor.org/stable/4603564>

Accessed: 20/06/2014 12:21

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at
<http://www.jstor.org/page/info/about/policies/terms.jsp>

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.



Springer is collaborating with JSTOR to digitize, preserve and extend access to *Human Ecology*.

<http://www.jstor.org>

Coastal Resource Management and Social Differences in Philippine Fishing Communities

James F. Eder¹

Class, ethnic, and gender differences among fisherfolk powerfully influence how the benefits and costs of coastal resource management programs are perceived and experienced in Philippine fishing communities. These and other social differences also limit the efficacy of community participation in these programs and hence the role that local communities can be expected to play in fisheries co-management regimes. Greater attention to institutional changes is needed if the management potential of such regimes is to be realized.

KEY WORDS: fisherfolk; coastal resources; community; co-management; Philippines.

INTRODUCTION

This paper considers how an effort to institutionalize coastal resource management at the municipal level in the Philippines has unfolded against the characteristic class, ethnic, and gender divisions of lowland Philippine society. It draws inspiration from recent suggestions that the study of community-based natural resource management would be better served by focusing on institutions rather than on community *per se* (Agrawal and Gibson, 2001; King and Durrenberger, 2000) and, more particularly, on how institutions shape (and are shaped by) the multiple interests and actors within communities (Agrawal and Gibson, 1999, p. 640). Of special interest here is how institutional arrangements may influence both the reality and the perception of local community “participation,” that much sought-after element of resource management projects in general.

¹Department of Anthropology, Arizona State University, Tempe, Arizona 85287-2402, 480-965-5530; e-mail: James.Eder@asu.edu.

The particular effort I shall examine, the Coastal Resource Management Project (CRMP) jointly sponsored by USAID and the Philippine government, is one of many such projects that came about in the Philippines and elsewhere in Southeast Asia following growing realization that fisheries resource over-exploitation and coastal environmental degradation have complex social, economic, and political origins, and that coastal resource management issues will only be effectively addressed when fishers and other resource stakeholders are more involved in the management process (Pomeroy, 1995). Put differently, whatever the laws and regulations might be, fisheries cannot be managed effectively without the cooperation of fishers to make those laws and regulations work (Pomeroy, 1995, p. 144).

Putting this now well-established wisdom into practice in the Philippines, however, has proven a challenging task. The basic approach appears sound enough: higher levels of government are expected to provide the regulatory and infrastructural capabilities that local fishing communities lack, while the fishing residents of these communities are in turn expected to contribute their greater local knowledge to management initiatives and to mobilize one another in support of those initiatives (Russell and Alexander, 2000, p. 36). A number of projects structured along such “co-management” lines have now been implemented in various parts of the country. While I believe that the jury is still very much out regarding their success, a study of one such project (Sunderlin and Gorospe, 1997) concluded that local social stratification (among other things) undermined a rather substantial fisheries co-management initiative in San Miguel Bay. More generally, a recent review of experience to date in the Philippines and elsewhere with community-based marine protected areas (often the centerpiece of co-managed CRM projects) found that such social and economic conditions as poverty and lack of political will hinder implementation of the broader management policies needed for such community-level initiatives to succeed (Christie *et al.*, 2003, p. 23). For these and other reasons, White *et al.* (2002) argue that in the Philippines, in addition to local communities and other stakeholders, “local municipal governments must play a key role from the beginning” if CRM projects are to be more effective than they are at present (2002, p. 22).

These findings helped focus my own interest in the CRMP and what I take to be a sort of local-level experiment in institution building that attempts to fold a new activity (coastal resource management) into an existing institution (municipal government). Specifically, I shall explore in this paper how the CRMP has unfolded in the context of municipal governance, mayoral politics, and the characteristic economic and social diversity of frontier Philippine communities. The project has, to date, seen aggressive enforcement of existing laws protecting municipal fishing waters from intrusions by

large-scale commercial fishing operators; prohibition of compressor-aided spear fishing and confiscation of all privately-owned compressors in the municipality (compressors are also closely linked to the use of cyanide); demarcation of a series of proposed marine protected areas; and an extensive public education campaign preparatory to an intended crackdown on certain fishing practices, such as beach seining and the use of fine-mesh nets, which are technically illegal but still commonly employed.

My purpose here, however, is not to ask whether or to what degree these various CRMP initiatives have led to reduced exploitative pressures on the hoped for recovery of municipal fish stocks (although there is some encouraging anecdotal evidence that the project has indeed enjoyed some success in these regards). Rather, I take some near term reduction in the total fish catch due to CRMP efforts as given and ask which particular fishers are catching or are likely to catch fewer fish (or now will have to work longer or harder to catch the same amount of fish), as a result of these initiatives. Hence my underlying interest here in what I sometimes think of as the “burdens of conservation” and how these burdens come to be shared among the various kinds of fishing households found in coastal Philippine fishing communities. Put differently, *some* reefs will be protected and *some* ecologically-undesirable fishing practices will be eliminated, and so on. But how are decisions in these regards made, and what political forces and notions of social good do these decisions reflect?

I shall show that while local residents are generally agreed that depletion of fish stocks in municipal waters due to illegal and destructive fishing methods and growing numbers of fishers is a matter needing government action, they disagree regarding what, precisely, most needs to be done, in ways that track the social divisions of lowland Philippine society and vary with the degree and kind of household dependence on coastal resources. This paper shows how ethnicity, class, and gender each influence local understandings of coastal resource issues, and it considers how the burdens of resource-conserving measures as implemented or called for by this particular project are in turn distributed unequally across class, ethnic, and gender lines—the principal structural fault lines of lowland Philippine society. My findings suggest that in the real world of coastal fishing communities, greater attention is needed to the intermediate and sometimes conflicting notions of societal good that may lie between the polar notions of individual good and common good.

These observations, I argue, illuminate two central but deeply problematic notions associated with both the theory and the policy of coastal resource management, “community” and “participation.” As a government project aimed primarily at the municipal level but with some local-level community involvement, the CRMP combined elements of both

“top-down” and bottom-up” approaches. While this was one of the features that made it attractive for study, the CRMP was in some ways less participatory than many of the NGO-sponsored coastal resource management projects, in Palawan and elsewhere in the Philippines, that emphasize community organizing and community empowerment and otherwise pride themselves on the “bottom-up” nature of their approach. In my concluding discussion and drawing on comparative observations from other CRM projects, I argue that whatever the approach, the same social cleavages in lowland Philippine society that lead to unequal sharing of the burdens of coastal resource conservation also render “community participation” a subjective and easily politicized notion. I also argue that the task of institutionalizing a co-management approach to CRM, whereby management responsibility is shared between local fishing communities and various levels of government, will require greater attention to institutional change than it presently receives if CRM programs are to take adequate account of the full range of local interests and concerns.

THE RESEARCH SETTING

The Ethnographic Setting

The municipality of San Vicente lies on the northwest coast of Palawan Island facing the South China Sea. It consists of 10 communities or *barangays* (the lowest level governmental unit in the Philippine political system), one of which is also the municipal seat. San Vicente’s total land area is 840,000 hectares and its population (according to the 2000 census) is 21,654 persons.

San Vicente’s 10 communities lie in a row along the coast, such that each community includes a stretch of coastline; a corresponding area of ocean out to the 15-km municipal waters limit; and a corresponding area of land, the coastal portion of which is generally flat and cleared for settlement and the interior portion of which is mountainous and forested. This geography suggests the characteristic mixed “fishing and farming” economic base of San Vicente communities: fishing with various kinds of equipment and at various levels of capitalization, along the coast; irrigated rice agriculture on those areas of the coastal plain favored by adequate water resources; and a mixed upland agriculture—corn, root crops, coconuts, bananas, and fruit trees—elsewhere on the coastal plain and in the foothills.

Commercial logging was once also of great importance to San Vicente’s economy. When a total logging ban was proclaimed in Palawan in 1992, a local logging concession was the municipality’s largest employer;

when the logging operation was closed, many former employees became fishermen. While a variety of agricultural products reach external markets, the municipality's most visible market links today concern fishing, and fishing boats and fisherfolk are an omnipresent feature of everyday coastal life. Most fishers operate motorized boats and employ small-scale, artisan gears, mainly gill nets of various kinds, but also throw nets and drift nets. Hook-and-line fishing is also common, particularly among fishers with small, oar-powered boats. Fresh and dried fish of all kinds is trucked daily to the east coast of Palawan for sale in Puerto Princesa City and for transshipment to Manila. Special arrangements also govern the marketing of squid (a seasonally-important San Vicente product) and, more recently, seaweed (farmed by some fishing households seeking an alternative livelihood in the face of coastal resource depletion). At higher levels of capitalization, a substantial pearl farm operates in municipal waters, and several times weekly a small plane arrives to transport live fish, principally leopard grouper raised in fish cages or on-shore "aquariums," bound for restaurants in Hong Kong and Taiwan.

The relative contributions of agriculture, fishing, and other economic activities to household livelihoods vary widely, within and between communities and households (Eder, 2003). Ecology explains some of this variation; not all communities are equally blessed with good fishing grounds or with land suitable for agriculture. Early attempts at coastal resource management by local San Vicente officials may also have contributed to the municipality's varied subsistence economy. Of greater interest here, however, are the considerable ethnic differences between San Vicente's residents, differences that also provide an important window on the complexity of municipal economic life. An oft-heard anecdote in one fishing-oriented community of 200 households, for example, is as one walks along the beach one passes from Bikol, to Masbate, to Samar, and then on to Cuyo, all various distinctively different ethnic points in the Philippine archipelago.

How such ethnic variation came to be is wrapped up with the history of San Vicente's settlement, itself part of the larger story of the settlement of Palawan Island generally (see Eder and Fernandez, 1996). But for purposes here, most of San Vicente's inhabitants can be envisioned as falling into three categories, based on a timeline of settlement. The first category consists of Tagbanua, members of a congeries of people indigenous to Palawan who identify locally as *Tandolanen* Tagbanua and who claim to be linguistically and culturally distinct from the anthropologically better-known Tagbanua of the Aborlan region (Fox, 1982). Tagbanua in San Vicente are today extensively acculturated to and intermarried with Cuyonon and other lowland Filipino peoples, but they continue to self-identify as an indigenous

people, and as the region's original inhabitants, they may be referred to by others (often condescendingly) as "natives" or as *naturales*. Tagbanua in San Vicente today mostly engage in shifting cultivation, wage labor, and the collection and sale of non-timber forest products.

The second category consists of Cuyonon and Agutaynen, peoples who originated elsewhere in the Palawan region (on small islands lying between Palawan Island and Panay Island) and who characteristically combine an upland mixed farming or irrigated rice lifeway with part-time dependence upon fishing. Cuyonon and Agutaynen migrants first settled in San Vicente during the 1950s and (as elsewhere on Palawan Island) were responsible for most of the pioneer forest clearance on the coastal plain. The third category of local inhabitants consists of migrants from still more distant locales in the Philippines, particularly "Visayans," as Cebuanos, Warays, Masbateños, and other migrants from the central Philippines are known locally, but also migrants from Mindoro, Bikol, and from Cebuano-speaking parts of Mindanao. These migrants began reaching San Vicente during the 1960s and 1970s and largely fish for a living.

Such unidimensional, thumbnail characterizations of San Vicente's residents, of course, do not do justice to the considerable cultural similarities and differences within and between these ethnic categories. Some Visayans primarily farm and some Cuyonon primarily fish for a living, and so on, and class and other forms of difference are at play as well. And neither do these brief characterizations tell us whether such systematic associations between economic lifeways and ethnic backgrounds as are present owe primarily to some cultural imperative (it is thought locally, for example, that Visayans are naturally attracted to the ocean, just as Cuyonon are to the land) or are better attributed to time of arrival (Cuyonon migrants generally preceded other migrants) or to some other material circumstance. But for now I simply want to establish that San Vicente's coastal communities are marked by considerable ethnolinguistic and ecological variability, such that the nature of dependence upon coastal resources varies widely, within and between communities.

The Institutional Framework

The Philippines has a long history of government regulation of the nation's fisheries, and whatever folk notions of resource management or property rights that might have existed in the nation's coastal areas have long since weakened or disappeared in the face of technological change and outside political and economic forces (Pomeroy, 1995, p. 145; Russell and Alexander, 2000, p. 38). This is more true in Palawan, where many fishing communities are recently founded and composed of migrants of diverse

origins, whose abilities to establish and enforce norms of fishing behavior through collaborative processes of decision and rule-making are, at best, weakly developed. Hence the institutions, both old and new, that figure in this paper are largely of the more formal, governmental sort.

The national government's own responsibilities for fisheries management were most recently delineated in the National Fisheries Code of 1998 (Republic Act No. 8550), and municipal government (rather than local community) authority over local fishing grounds dates back to the colonial period (Pomeroy, 1995, p. 151).² But ineffectual regulatory measures and lack of enforcement, destructive fishing practices, illegal commercial fishing in municipal waters, and population pressure have brought Philippine coastal ecosystems to their current diminished state. Coastal resources are seriously depleted, environmental damage is widespread, and poverty levels among municipal fishers are among the highest in the country.

Since the passage of the Local Government Code of 1991, which devolved greater control over coastal resource access to local levels through policy and institutional reforms, a wide variety of government and NGO "co-management" initiatives have attempted to partner various levels of government with local fishing communities in more active and participatory forms of coastal resource management (Pomeroy and Pido, 1995). A growing academic literature considers the successes and failures of these various initiatives; see, e.g., Christie *et al.* (1994); Pomeroy and Pido (1995); Sunderlin and Gorospe (1997); White *et al.* (2002); and the collections of case studies in White *et al.* (White *et al.*, 1994, pp. 97–98) and Rivera and Newkirk (1997).

Meanwhile and in part due to the very success of some of these reforms and initiatives, the formal institutional landscape governing CRM programs is muddled. Not only are there various levels to this institutional landscape (Pinto, 2000), including numerous government agencies, NGOs, and donor-assisted and government programs (see White *et al.*, 2002, p. 8), but institutions at different levels (or even at the same level) may come into conflict due to contradictory policies or to overlapping or ambiguous jurisdictional boundaries. In these circumstances, relations of power and other aspects of local political ecology loom large as strategically placed actors exploit to

²In the case of Palawan, natural resource management schemes such as the CRMP also unfold in the context of a distinctive provincial institutional framework, centered on Palawan's Strategic Environmental Plan (SEP). Passed into national law in 1992, SEP provides for the eventual zoning of all of Palawan in terms of the Environmentally Critical Areas Network (ECAN), intended to provide for controlled, environmentally-sound development on land and in coastal areas (Austin, 2003, p. 126). Also established by SEP and overseeing development in the province of Palawan is the Palawan Council for Sustainable Development (PCSD), composed of political appointees and permanent staff members collectively responsible for development planning.

their advantage local policy and jurisdictional uncertainties (see Lawrence and Bryant, 2003).

A seven-year (1996–2002) joint project of USAID and the Philippine government, the CRMP provided technical assistance and training to coastal communities, local government units, non-governmental organizations, and national government agencies to improve management of coastal resources. At the national level, the Department of Environment and Natural Resources (DENR) was the lead implementing agency, but numerous collaborative relationships were formed with other departments and agencies as well. Nationally, CRMP aimed to assist the Philippine government in finding practical solutions to four key CRM problems: jurisdictional issues, mangrove management, commercial fisheries management, and biodiversity conservation. Locally—the aspect of the CRMP of greatest interest here—the project aimed to help communities and municipal governments institutionalize CRM planning and implementation. The CRMP unfolded at six local “learning areas” throughout the central and southern Philippines; that San Vicente is one of these six areas was one of the reasons I selected it for study.

San Vicente’s own CRM experience, however, predated the arrival of the CRMP—indeed, it was among the reasons that CRMP personnel selected it as a learning area. The municipality’s coastline, by one estimate approximately 120 km, if the coastlines of the various small islands that help compose San Vicente are included, certainly provides an appropriate locale both for coastal resource exploitation and for attempts to better manage that exploitation. The northwest coast of Palawan generally is one of the nation’s major fishing grounds (Malampaya Sound, well known in the Philippine fishing industry, lies just to the north of San Vicente), and in decades past fish, squid, crabs, and other marine products were obtained in abundance in San Vicente waters. Since the 1970s, as local coastal populations of migrant fisherfolk have continued to grow, and as visitation by large-scale commercial fishing outfits based elsewhere has become ever more frequent, local fish catches (as measured in kilograms of fish per fisher per day) have steadily—even precipitously—declined (Arquiza, 1999, pp. 114–115), even as local fisherfolk have turned to ever more intensive and efficient capture techniques.

The resulting litany of management problems is recognizable by anyone familiar with CRM problems elsewhere in the Philippines: blast fishing; use of sodium cyanide to capture live food fish; use of beach seines and other drag nets; use of fine-mesh nets; encroachment by commercial fishing vessels; use of Danish seines; encroachment of tourist facilities and resulting problems (littering, discharge of waste, boats anchoring on reefs, and so forth); increased sediment load due to inappropriate agricultural

practices, logging, and local residential development; and so on (Arquiza, 1999; Municipality of San Vicente, 2001).

These were the issues that the CRMP grappled with during its 7-year tenure in San Vicente. In that grappling it joined forces with San Vicente's municipal government, on the one hand assimilating some of the municipality's own prior CRM initiatives and, on the other, attempting to prepare the municipal government (in particular, the Municipal Planning and Development Office, or MPDO) to adopt and implement the CRMP's own larger and more ambitious vision of CRM, once the CRMP itself was phased out. Sometimes working independently and sometimes together with municipal government employees, and in addition to the earlier-noted CRMP initiatives, CRMP personnel organized numerous public meetings, designed and implemented various alternative livelihood projects, and assisted the MPDO in the preparation of a 5-year (2001–2005) municipal coastal resource management plan (Municipality of San Vicente, 2001).

The CRMP ended in mid-2002 with a carefully orchestrated turnover of CRM plans and responsibilities to local government agencies. CRM efforts in San Vincente were off to a promising start but with two significant budget issues unresolved: whether the mayor would or could create a full-time staff position in the municipal government dedicated to CRM implementation, and where local communities could obtain the funding needed to monitor effectively the marine sanctuaries now being created. In addition, several substantive matters required attention, in particular the need for a zoning system to govern the locations of the growing numbers of seaweed farms and various other marine commercial enterprises.

CRMP INITIATIVES AND SOCIAL DIVISIONS IN THE LOWLAND PHILIPPINES

I now consider how the political realities of class, ethnicity, and gender—some of the principal social divisions of lowland Philippine life—affect how local community members perceive CRMP initiatives and how, in turn, community members (acting on these perceptions) have influenced their design and implementation. I thus deploy an understanding of communities as sites of pluralism rather than of social homogeneity, an understanding that runs counter to prevailing management orthodoxies and to the idealized imagery of local communities these orthodoxies typically embody (Natcher and Hickey, 2002, pp. 350–351). More specifically, I illustrate what Christie *et al.* (2003, p. 22) characterize as inadequately understood and “frequently contentious” social interactions at various levels and across

gender, class, and ethnic lines may negatively impact community-based efforts to implement CRM projects.

I imagine CRM projects in general as asking fisherfolk to curtail certain fishing practices that have proven economically attractive (or necessary) but which have been judged environmentally undesirable, i.e., fishers are asked to make near-term sacrifices for the longer-term benefit of all. But probably no such project—given pre-existing variability in fishing practices, levels of capitalization, and the role of fishing in household economic life—can ensure (even if it wanted to) that the burdens of sacrifice are distributed equally among all community members. Hence my basic question here: how are these burdens perceived and experienced by different social actors under this particular CRM project?

By highlighting some of the ways that structural variables such as class and ethnicity can shape and restrict collective action in relation to resource use (see Johnson, 2001), I aim to illustrate the inexorably political nature of even the seemingly most straightforward resource use policy measures. More generally, as regards the dilemma that natural resource management poses for collective action, I shall argue that the ways in which individuals understand and respond to the costs and benefits of existing and potential institutional arrangements (Baland and Platteau, 1996; Johnson, 2001, p. 95) are powerfully mediated by their positions on a grid of class, ethnic, and gender differences, each with its own implications for collective action. I consider in turn CRMP efforts to restrict illegal commercial fishing in municipal waters; to end compressor-aided fishing (a form of fishing closely linked with the use of cyanide); to create a series of marine protected areas; and to address a variety of gear issues, particularly the use of beach seines.

Illegal Commercial Fishing in Municipal Waters

One of the most difficult and widespread coastal resource management issues in the Philippines concerns the illegal encroachment on municipal waters of commercial fishing boats using trawls and other highly efficient gear (see, e.g., Russell and Alexander, 2000; Sunderlin and Gorospe, 1977). As elsewhere in Southeast Asia (see, e.g., Lowe, 2000), local patterns of resource use (and misuse) are inextricably bound up with a larger political economy marked (in the local view) by a class-based social division between mostly poor and otherwise “small” local fisherfolk and wealthy and politically influential “outsiders” based, in the case of San Vicente, in or near Manila. While Republic Act No. 8550 (1998) reaffirmed the long-standing and widely known 15-km municipal fishing waters boundary in the Philippines, intrusions of commercial trawlers into municipal waters are

commonplace and are often said to occur with the collusion of municipal mayors who (it is widely assumed) likely receive some sort of payoff from commercial fishers in return for looking the other way.

When the present mayor of San Vicente took office, one of his first big political moves was to vigorously pursue commercial trawlers that entered local waters. A speedboat was acquired, a system of telescopes was set up (to triangulate on the positions of suspected intruders) and, with some regularity, boats have been boarded and seized and their captains brought to Puerto Princesa City to be charged in provincial courts. Whether this pattern of aggressive prosecution will continue remains to be seen, and even at present the mayor's political opponents suspect his motives are not as virtuous as this brief account may make them to appear. But the mayor is, after all, only enforcing the law, and the fishers I spoke with were generally agreed that illegal commercial fishing was less a problem now than in the past. Given that such fishing in neighboring municipalities is reported to continue unabated (Taytay and Cuyo are often singled out in this regard, and the mayors of those municipalities are widely reported to condone or even to have business involvements in illegal fishing) and, more generally, that lack of political will on the part of local officials (Arquiza, 1999, p. 109) is so commonly cited as a factor in the persistence of illegal fishing practices of all kinds, that even one mayor in Palawan has apparently found such will is interesting and (I would like to believe) encouraging.

Compressor-Aided Fishing

The town mayor made a second dramatic political move when he ordered the confiscation of all compressor rigs in the municipality. (Despite denials from their owners, use of compressors is closely linked to the use of cyanide to capture live fish; Lowe, 2000, p. 249). Nearly 30 compressors were confiscated from one community alone; their owners received some direct compensation and were also invited to participate in a 2-week livelihood training program with a daily honorarium. While some persons suffered significant economic losses, the great majority of fishers applauded the move, and as with his willingness to vigorously prosecute illegal commercial fishing, the mayor (now given to claiming that San Vicente owns more compressors than any other municipality in the country) created significant political support for other elements of the municipality's CRM program.

Nonetheless, cracking down on the clearly illegal and much decried activities of outside commercial fishing boat operators and of a small minority of local residents (those presumed to be engaged in cyanide and blast fishing) is in some ways easier politically than is implementation of

the balance of the CRMP agenda. Indeed, many fisherfolk I spoke with were of the opinion that the municipality's CRM agenda should be limited to just that: crack down on the illegal activities, and let the ordinary person get on with the business of making a living. The problem posed by this view, of course, is that with too many people chasing dwindling numbers of fish, still other conservation measures are needed, but there is less political agreement regarding which particular measures and how they should be implemented.

Marine Protected Areas

The basic notion behind marine sanctuaries is to rehabilitate habitat, conserve biodiversity, and increase fisheries production by designating certain suitable marine areas as off-limits for human entry and exploitation. While marine sanctuaries may also be established to protect mangroves and sea grass beds, in the Philippines they are most commonly and visibly associated with the protection of coral reefs, and that is their primary purpose under the CRMP. The particular issue I wish to explore here concerns the process whereby particular coral reefs come to be designated for protection, because while the CRMP called upon each local community in San Vicente to select one or two such reefs, it was left to the communities themselves to determine which ones, with the understanding that once sites for protection were selected, the CRMP and municipal officials would follow up with the necessary technical and legal assistance.

Writing of local fisherfolk perceptions of marine sanctuary proposals in the Honda Bay region elsewhere in Palawan, Austin observes that while conventional coastal management understandings suggest that the most suitable candidates for sanctuary status are coral reefs that are still primarily intact, some local fisherfolk (and some local environmentalists) prefer sites that are already partially degraded, in order to allow them to regenerate (Austin, 2003, pp. 297–298). A local view of marine sanctuaries as primarily about food production (rather than biodiversity conservation) is one likely explanation for such a preference (see Austin, 2003, p. 293); another is ease of surveillance and understandable local concern that distant sites cannot be effectively monitored against intrusion (see Christie *et al.*, 1994, p. 110). Both issues certainly figured in community discussions of prospective sanctuary sites in San Vicente. While community selection and official (municipal) designation of Marine Protected Areas (or MPAs, as marine sanctuaries are known under the CRMP) is still an on-going process, those sites that have been selected to date all center on close-in, partially-to-seriously degraded coral reefs, rather than on more distant and more pristine reefs.

This series of decisions highlighted an important local class difference between those fishers owning motorized pumpboats (and hence able to reach relatively distant fishing grounds) and those fishers owning only smaller, oar-propelled outriggers (who hence relied on the closer-in fishing areas). The decisions themselves were unsurprising; in all local communities in San Vicente the majority of fisherfolk own motorized pumpboats and do not exploit the closest reefs anyway, as these reefs have long since been depleted of fish. But the outcome, of course, was that those fishers who, for whatever reason (usually relative poverty), relied on oar-power to go fishing—and hence on these close-in reefs—now found themselves greatly disadvantaged by the prospect of greater travel times.

Growing public awareness of this outcome left the municipal government and the CRMP vulnerable to the politically dangerous charge that their conservation efforts were hurting the very people that government should endeavor to assist, a charge soon found appealing by the mayor's political opponents. Along these lines, I learned of a case where a local community resident who had initially supported the establishment of nearby MPAs later reversed his position, because *nagalit na 'yong mga kamaganak ko sa akin*, "my (poorer) relatives (i.e., those without motorized pumpboats) got mad at me." For their part, CRMP personnel well understand that some fisherfolk, often including those most poorly-off economically, will necessarily be disadvantaged or even displaced by successfully-implemented CRM measures. Hence the emphasis in all CRM projects in the Philippines on alternative livelihood training; the CRMP's own efforts in this regard centered on seaweed farming and the formation of cooperatives to produce coconut vinegar and bottled sardines. For now, and leaving aside matters of equity, most residents seem at best ambivalent about marine sanctuaries (see Cañete, 2000, p. 200); many like the idea in principle but are concerned in practice about community ability to effectively monitor the sanctuaries (in part a budgetary issue) and to discipline those inevitable "others" who would attempt to continue to fish in them.

Gear Issues

While gear conflicts in Philippine capture fisheries primarily reflecting class differences in wider Philippine society (e.g., the earlier-noted conflicts between small-scale municipal fishers and more highly capitalized commercial fishers or, at a more local level, between fishers with motorized boats and those without), in San Vicente gear conflicts have ethnic dimensions as well. Thus local perceptions of the CRMP's efforts to establish marine sanctuaries were in part ethnicized, reflecting a characteristic

difference between Visayan and Cuyonon capture techniques. Most residents of Visayan origin who fish typically derive their primary livelihoods from fishing and employ motorized pumpboats and various kinds of nets. While most local Cuyonon residents are primarily farmers, some fish on a part-time basis with a hook-and-line. Several Cuyonon farmers who fished from small outriggers in this “on the side” fashion resented the establishment of the close-in MPAs, which they saw as unnecessarily restrictive. In their view, only the nets and other activities of Visayan fisherfolk do damage to coral reefs, so why not prohibit net fishing in the proposed MPAs, and let hook-and-line fishing continue as before. (Because the interests of the majority of Visayan fishers who own motorized pumpboats prevailed in the siting of MPAs, the fact that their nets are excluded from the nearby, degraded reefs is no hardship for them.)

Cuyonon perspectives such as these are one element of a wider local discourse that attributes much of the degradation of Palawan’s fisheries to the various and increasingly efficient fish capture techniques introduced there by migrant Visayan fisherfolk; e.g., *hulbot-hulbot* or Danish seine, a destructive fishing method that employs large weights and scare-lines. Considering the post-1950 settlement of the Palawan fishing municipality of Panacan by migrants of Visayan origin, Seki (2000, pp. 272–273) describes the increasingly efficient fishing methods that successive waves of new arrivals introduced to the area—methods that, one presumes, were soon taken up by local residents.

My main concern here, however, is with beach seines, a capture technique using a long, fine-mesh net set from a boat but hauled from the shore by two teams, one at each wing of the net, composed mainly of women and children. (For detailed descriptions of beach seines and beach seining in the Philippines, see Yano, 1994, pp. 26–30, and Cañete, 2000, pp. 164–173). Use of beach seines in the Philippines is restricted by law and, in San Vicente, is banned by municipal ordinance and the municipality’s own CRMP-inspired coastal resource management plan. Although the ban is not presently enforced, there is general agreement among coastal resource management experts that beach seining leads to degradation of sea grass beds, undesirable physical alteration of near-shore sea bottoms, and a high incidence and waste of by-catch.

Beach seining, however, has considerable populist appeal, and it is an important secondary (or even primary) occupation of many women and children from economically disadvantaged coastal households. There is hence a significant gender dimension to beach seining, and in a coastal economy where men dominate the capture and, to an important degree, even the marketing of fish, female beach seiners, working either as net haulers or net owners, earn incomes from fishing relatively free of the direct control or

supervision of husbands or other males. Beach seining loomed particularly large in the household economies of female-headed coastal households, and in one San Vicente community where farming predominated and beach seining was the most common fishing activity, nearly half of women surveyed reported beach seining as either a primary or secondary occupation. In consequence, any attempt by the mayor to enforce the ban on this activity is likely to prove politically costly, as his political opponents would likely take up the same theme noted earlier, that the mayor shouldn't hurt "the small people" (*ang mga maliliit*) (who, after all, are just trying to survive) in his zeal to save the environment.

Differing local perceptions of beach seining, finally, track the same ethnic division noted above between Visayan and Cuyonon residents, although now with the tables turned. Visayan fishers I spoke with wanted to see the practice eliminated, arguing (as environmentalists do) that beach seining captures juvenile fish that would otherwise grow bigger and move out to sea. Cuyonon and Agutaynen farmers, in contrast, many of whom profited as hog raisers from beach seining (see below) even if they did not engage in it themselves, defended beach seining as a traditional and non-destructive practice (see Christie *et al.*, 1994, p. 110) and felt that the environmentalist arguments against it were misguided. Indeed, some joined a vigorous counterdiscourse about beach seining, variously arguing that the shallow ocean waters near the beach are not a "nursery" and that the small fish captured there are not of the sort that grow bigger anyway; that the fish caught with beach seines do not ordinarily live near the beach but only approach it during certain climatic conditions (a line of reasoning that explains why beach seiners sometimes capture few fish); and that periodically stirring up the ocean floor near the beach is good for the marine environment.

DISCUSSION

The positive role that various sorts of co-management arrangements might play in fisheries management has received considerable scholarly and policy attention in recent years (Jentoft *et al.*, 1998). The basic notion behind co-management is cooperative sharing of coastal resource management responsibilities between local groups or communities of fisherfolk and various levels of government (King and Durrenberger, 2000, pp. 11–12; Pinkerton, 1989). According to McCay, co-management refers to:

the democratization of fisheries management such that resource users have a strong say in developing the goals and rules of management. It therefore implies that the goals and needs of resource users—and by extension, their communities—are taken into account (McCay, 2000, p. 210).

Co-management was not explicitly referenced by CRMP personnel or in CRMP literature as a project goal. But both it and the other CRM projects I observed in Palawan have certainly aimed to increase local user participation in the management process, with the expectation that such participation would lead to more effective solutions to coastal resource management problems and would otherwise improve the institutional structure of governance (McCay and Jentoft, 1996, pp. 237–238), and it seems reasonable to consider these projects in this perspective.

At a minimum, a viable coastal resource co-management arrangement would seem to require, on the local side, (1) the presence of appropriate local individuals able and willing to *participate*, and (2) the presence of suitable *community-level institutions*, to mobilize that participation and ensure that it “matters.” Both “community” and “participation,” of course, are problematic enough notions in the abstract. But in present-day coastal Philippine communities they become more problematic still, for the same sorts of social differences that affect how the burdens of conservation are perceived and experienced also powerfully influence how communities are constituted for CRM purposes to begin with, how institutional arrangements actually work in practice, and how the subsequent participation of differently situated individuals takes place.

What communities “are” or should best consist of for CRM purposes is a much-discussed topic (see McCay, 2000, pp. 208–210; Zerner, 2000, p. 12). Helpful here is the distinction between communities of interest or “functional communities,” such as fishers’ organizations of various kinds, and communities of place or “territorial communities,” such as the local *barangay* government units that together compose the municipality of San Vicente (Jentoft *et al.*, 1998, p. 429). In the event, the CRMP worked directly or (through the apparatus of the municipal government) indirectly with communities of both kinds, and I suspect this is typical of CRM projects in the Philippines generally. Where the aim is simply to consult with or nominally involve as many people as possible, and to ensure representation of as many different interests as possible, this is a workable enough strategy. But where the aim is to *institutionalize* coastal resource management, such an ecumenical approach may be less helpful.³ Thus

³The institutionalization of resource management has different dimensions in those coastal regions of the world where customary systems of marine tenure still operate (e.g., the Solomon Islands; see Aswani, 1999, and Hviding, 1996). In these latter settings, calls to “institutionalize” coastal resource management in the face of intensified maritime commercial activities refer, in part, to the proposed codification and inclusion of “traditional” systems of marine tenure in modern systems of fisheries management (Aswani, 1999, pp. 449–450). Again, coastal resource managers in the Philippines lack the potentially useful building blocks of customary marine tenure systems and well-developed local norms of fishing behavior as they attempt to institutionalize CRM programs at the community and municipal levels.

CRMP personnel had numerous conversations with and encouraged the participation of fisherfolk and fishers' organizations, but because CRMP's official mandate was to institutionalize CRM at the municipal level of government such "community participation" as the CRMP called forth was ultimately assimilated to the barangay-municipality-state political grid—i.e., to the grid of government-mandated territorial communities. As a result, the customary understandings and participatory efforts of various *other* kinds of communities, be these the self-consciously organized functional communities such as fishing boat operators associations, or the unorganized "communities of interest" that track the various social divisions discussed earlier, had to be mapped on to the barangay-municipality-state system of governance and hence to the politics of that system.

Granted, there is considerable virtue to be had in drawing territorial communities into co-management arrangements. Particularly in coastal communities where (as in Palawan) fishing is typically only one of several livelihood strategies, working with local communities of place promises to better represent the interests of those many residents who are not fishers themselves but who depend indirectly on the well-being of coastal resources (Jentoft *et al.*, 1998, p. 429–430; Wingard, 2000, p. 53). But whatever other economic and social characteristics a Philippine barangay may have, it is also a unit of local government and inexorably *part* of the state—and hence should not be too easily imagined as having sufficient autonomy so as to "cooperate" with, much less to bargain with, the state (Li, 2001, p. 164). And if this is true of the local barangay community, the more so of the municipality. Granted, the municipality can in some ways be imagined as a kind of "greater community" (this is apparently how the CRMP imagined San Vicente), but it is vested with considerably more power and resources than a barangay and in important ways functions as a kind of mini-state, with (among other things) the municipal mayor in command of an extensive system of patronage.

This line of reasoning lies behind some of the current skepticism regarding whether local communities actually become empowered by community-based CRM programs in ways that will allow the members of those communities to gain control over local resources and the benefits that flow from them (Austin, 2003, pp. 47–49). Certainly the doubts of some social scientists in these regards are matched by uncertainty (and, at times, outright confusion) among the community members themselves, regarding the degree to which they as individuals are actually participants rather than bystanders.

Closely linked with the problematic understanding of community is the problematic understanding of participation. Not surprisingly, fisherfolk with reservations about particular CRM measures or proposals do not

usually say “I’m going to be hurt by that”; rather, they say “*we’re* going to be hurt by that,” implicitly referencing other, presumably like-minded people—hence my emphasis on social divisions and positionality. But in the case considered here, such fisherfolk—those with CRM-related objections or concerns—also sometimes claimed that they had not been given adequate opportunity to *express* their objections and concerns. That is, they claimed that they had not *participated* in the relevant decision-making process, even though (as they themselves would point out) it was ostensibly participatory, and indeed, numerous local community meetings were held. Thus, regarding the siting of marine protected areas, fishers who relied on oar rather than on motor power did not simply feel that they had lost a vote; they felt that their voices and concerns had not received an adequate hearing.

All of the community meetings in question occurred prior to my field research, and I was hence unable to observe first-hand the decision-making processes that characterized them. But based on subsequent interviews with a variety of attendees with different involvements with and perspectives on the CRMP, two generalizations may be ventured. First, the meetings were only weakly participatory. Project and local government officials did listen to the various viewpoints and opinions expressed by community members, but some of the CRM policies later announced and implemented were apparently not based on any sort of community consensus about how best to proceed. It was this aspect of the manner in which the meetings were conducted that gave rise to the numerous complaints later that local residents had not really participated in the larger process, even if they had attended the meetings.⁴ Second, some local voices and perspectives were simply never heard at all. Certainly few of the many economically marginal women and children involved in beach seining—individuals generally excluded from formal community discussions by the manner in which structural variables, such as class and gender, shape or restrict collective action and by their own perceptions of their place within their communities (see Johnson, 2001, p. 971)—even attended the meetings. Female beach seiners emphasized to me (but evidently did not speak out at any meeting with CRMP project managers) that while seining might technically appear to be a “fishing” activity, it is inextricably bound up with the local farm economy. The many smaller fish caught by seining are primarily used to prepare hog meal for use or sale on nearby farms, just the sort of local interdependence of fishing and non-fishing livelihoods (Eder, 2003), and interrelatedness of

⁴Utting aptly characterizes the simplistic models of consultation and dialogue often associated with community-based conservation programs as “technocratic participation” for their failure to adequately engage broader community interests and the views and priorities of weaker social groups (2000, p. 194).

the economic lives of women and men (Haraldsdottir, 2000, p. 141), that CRM planners ought more to attend to.

There are, to be sure, considerable logistical difficulties in making the residents of an entire municipality feel they have meaningfully participated (or had the opportunity to participate) in discussions of proposed CRM initiatives. And an important cultural factor is operating here as well. Some of the complaints I heard from local residents that they had not been adequately consulted were strongly reminiscent of the traditional system of electoral politics, whereby candidates for office are expected to appeal to voters in person for their support. But whatever the reasons, that large numbers of local residents did not appear to feel participant in or ownership of the CRMP became, given the association between the project, the municipal government, and the persona of the mayor, easily exploitable by the mayor's political opponents, a circumstance that has threatened to undermine some of the project's substantial accomplishments. Put differently, for the many coastal residents for whom social justice is a matter of everyday concern, there was a "politics-as-usual" unfairness about the way the CRMP unfolded. Observations such as these help explain why issues of resource management generally have increasingly been linked to calls for greater social justice, both in the Philippines (Utting, 2000, pp. 189–194) and elsewhere (Christie *et al.*, 2003, p. 25).

Agarwal's typology of six different forms of participation in community-based forest management programs, ranging from "nominal" and "passive" participation to "active" and (ultimately) "empowering" participation (Agarwal, 2001, p. 1624) is helpful here. Much participation in CRM projects in Palawan, despite the ubiquitous "community-organizing" of government and NGO project sponsors, remains at the lower end of this scale, suggesting that the efficacy of community participation (Natcher and Hickey, 2002, p. 351) is still a matter in need of attention. And if this is important for community participation in general, it is more so, as Natcher and Hickey remind us, for the participation of women, the poor, and other local community members most at risk of subordination by local-level and possibly "top-down" political processes (2002, p. 351).

This, of course, is where new institutional arrangements are called for (Agrawal and Gibson, 2001). Some have called this "changing the rules" (Jentoft *et al.*, 1998, p. 427), and an emphasis on institutions focuses on the ability of communities to create and to enforce those rules (Agrawal and Gibson, 1999, p. 638) At the same time, however, co-management is itself

not so much about the rules per se as it is about the communicative and collaborative process through which these rules are formed: who participates, how debates are structured, how conflicts of interest are addressed, and how agreements are reached (Jentoft *et al.*, 1998, p. 427).

These are the elements that were to an important extent lacking in the CRMP initiative. As a government-sponsored project, it was perhaps to be expected that the CRMP would work primarily within the existing institutional environment. Indeed, in its carefully-crafted "PRO-CRM" plan to turn CRM responsibilities over to a long suite of existing municipal and provincial government agencies following its own termination, the CRMP made a great virtue of working in this fashion. But even CRM-involved NGOs, despite a certain amount of anti-establishment rhetoric, may display a similar zeal to work within existing institutional structures, in part perhaps to seek legitimation within those structures. Thus Haribon Palawan, a UNDP-funded NGO helping fishing communities on Palawan's Honda Bay to establish fish sanctuaries, helped set up formal "fish sanctuary management boards" whose memberships consisted of representatives from several local government units (e.g., the Bureau of Fisheries and Aquatic Resources); POs (peoples organizations, particularly local fishers associations); NGOs (including Haribon); "academe" (the local public school, and a provincial university); cooperatives; and the religious, youth, tribal, and private "sectors"—a virtual roll call of established political and social groupings in the rural Philippines.

Both the PRO-CRM and the sanctuary management board arrangements may have seemed, at the time, to be institutionally innovative but, with the wisdom of hindsight, they appear now more as old wine in a new bottle. Quite apart from the bureaucratic obstacles that such unwieldy arrangements would seem to pose to getting anything done at all, they seem guaranteed to ensure that the existing institutional inequities in wider Philippine society will continue to be projected on to local developments (see, e.g., Lawrence and Bryant, 2003). According to Natcher and Hickey,

despite enhanced local involvement in resource management, final decisions often remain reflective of only the dominant modes of prevailing power at the time, thereby muting alternative perspectives, insights, and systems of knowing. Consequently, by failing to account for community pluralism, local efforts all too often only soften the traditional top-down relationship long inherent in resource management, resulting in the continued subjugation of values and concerns of some community members (2002, p. 351).

In sum, perhaps too confident in their knowledge of where the collective coastal resource good lies, CRM project sponsors in Palawan have devoted insufficient effort to working with local people to determine where the greater social good lies, and to how existing institutional arrangements might best be strengthened or altered to promote social justice. In the absence of more strenuous efforts to channel greater power and authority to the weakest and most poorly represented local groups (Agrawal and Gibson, 1999, p. 641; Utting, 2000, p. 194), the efficacy of community

participation will remain limited, and effective co-management regimes will prove difficult to institutionalize. For the sorts of social differences and divisions discussed here will continue to demarcate intermediate and varying notions of social good, notions whose various adherents are unequally positioned to have their voices heard and their interests considered.

ACKNOWLEDGMENTS

Research in Palawan during January–April, 2002 was supported by a Fulbright research award and a sabbatical leave from Arizona State University (ASU). I am grateful to the Council for International Exchange of Scholars, the Philippine American Educational Foundation, and to ASU for this assistance. During June 2003, additional, comparative research in Palawan was made possible by an A. T. Steele Grant from ASU's Center for Asian Studies, for which I am similarly grateful. For their assistance during fieldwork I thank Meng Amihan, Regino Balofinos, Jovy Borres, Boy Magallanes, and Lilibeth Uapal. I also thank Rebecca Austin, Aloysius Cañete, Dante Dalabajan, Flora Leucadio, Melanie Hughes McDermott, and two anonymous reviewers for some helpful suggestions concerning the manuscript.

REFERENCES

- Agarwal, B. (2001). Participatory exclusions, community forestry, and gender: An analysis for South Asia and a conceptual framework. *World Development* 29: 1623–1648.
- Agrawal, A., and Gibson, C. C. (1999). Enchantment and disenchantment: The role of community in natural resource conservation. *World Development* 27: 629–649.
- Agrawal, A., and Gibson, C. C. (2001). The role of community in natural resource conservation. In Agrawal, A., and Gibson, C. C. (eds.), *Communities and the Environment: Ethnicity, Gender, and the State in Community-Based Conservation*, Rutgers University Press, New Brunswick, NJ, pp. 1–31.
- Arquiza, Y. D. (1999). *Rhythm of the Sea: Coastal Environmental Profile of San Vicente, Palawan*, Coastal Resource Management Project, Cebu City, Philippines.
- Aswani, S. (1999). Common property models of sea tenure: A case study from the Roviana and Vonavona Lagoons, New Georgia, Solomon Islands. *Human Ecology* 27(3): 417–453.
- Austin, R. L. (2003). *Environmental Movements and Fisherfolk Participation on a Coastal Frontier, Palawan Island, Philippines*, PhD Dissertation, Department of Anthropology, University of Georgia.
- Baland, J. M., and Platteau, J. P. (1996). The ambiguous impact of inequality on local resource management. *World Development* 27: 773–788.
- Cañete, A. M. L. (2000). From beach seining to *sapyaw* fishing: Innovation, competition, and conflict avoidance in a municipal fishery in central Philippines. *Philippine Quarterly of Culture and Society* 28: 158–223.
- Christie, P., McCay, B. J., Miller, M. L., Lowe, C., White, A. T., Stoffle, R., Fluharty, D. L., McManus, L. T., Chuenpagdee, R., Pomeroy, C., Suman, D. O., Blount, B. G., Huppert, D., Eisma, R. V., Oracion, E., Lowry, K., and Pollnac, R. B. (2003). Toward developing

- a complete understanding: A social science research agenda for marine protected areas. *Fisheries* 28: 22–26.
- Christie, P., White, A. T., and Buhat, D. (1994). Community-based coral reef management on San Salvador Island, the Philippines. *Society and Natural Resources* 7: 103–117.
- Eder, J. F. (2003). Of fishers and farmers: Ethnicity and resource use in coastal Palawan. *Philippine Quarterly of Culture and Society* 31: 207–225.
- Eder, J. F., and Fernandez, J. O. (1996). *Palawan at the Crossroads: Development and the Environment on a Philippine Frontier*, Ateneo de Manila University Press, Quezon City, Philippines.
- Fox, R. B. (1982). *Religion and society among the Tagbanuwa of Palawan Island, Philippines*. Monograph 9, National Museum, Manila.
- Haraldsdottir, G. (2000). Tradition, co-management, diversity, and the FAO in small-scale inland fisheries in Africa. In Durrenberger, E. P., and King, T. D. (eds.), *State and Community in Fisheries Management: Power, Policy, and Practice*, Bergin & Garvey, Westport, CT, pp. 131–148.
- Hviding, E. (1996). *Guardians of Morovo Lagoon: Practice, Place, and Politics in Maritime Melanesia*, Pacific Island Monograph Series 14, Center for Pacific Island Studies, University of Hawai'i Press, Honolulu.
- Jentoft, S., McCay, B. J., and Wilson, D. C. (1998). Social theory and fisheries co-management. *Marine Policy* 22: 423–436.
- Johnson, C. (2001). Community formation and fisheries conservation in southern Thailand. *Development and Change* 32: 951–974.
- King, T. D., and Durrenberger, E. P. (2000). Introduction. In Durrenberger, E. P., and King, T. D. (eds.), *State and Community in Fisheries Management: Power, Policy, and Practice*, Bergin & Garvey, Westport, CT, pp. 1–15.
- Lawrence, K., and Bryant, R. (2003). Changing local resource use practices: The role of community forestry policy. In *Paper Presented at the International Conference, "Local Land Use Strategies in a Globalizing World: Shaping Sustainable Social and Natural Environments"*, Institute of Geography, University of Copenhagen, Denmark, 21–23 August 2004.
- Li, T. M. (2001). Boundary work: Community, market, and state reconsidered. In Agrawal, A., and Gibson, C. C. (eds.), *Communities and the Environment: Ethnicity, Gender, and the State in Community-Based Conservation*, Rutgers University Press, New Brunswick, NJ, pp. 157–179.
- Lowe, C. (2000). Global markets, local injustice in Southeast Asian seas: The live fish trade and local fishers in the Togeian Islands of Sulawesi. In Zerner, C. (ed.), *People, Plants, and Justice: The Politics of Nature Conservation*, Columbia University Press, New York, pp. 234–258.
- McCay, B. J. (2000). Sea changes in fisheries policy: Contributions from anthropology. In Durrenberger, E. P., and King, T. D. (eds.), *State and Community in Fisheries Management: Power, Policy, and Practice*, Bergin & Garvey, Westport, CT, pp. 201–217.
- McCay, B. J., and Jentoft, S. (1996). From the bottom up: Participatory issues in fisheries management. *Society and Natural Resources* 9: 237–250.
- Municipality of San Vicente (2001). *Municipal Coastal Resource Management (CRM) Plan, 2001–2003*, Municipality of San Vicente, Palawan, Philippines.
- Natcher, D. C., and Hickey, C. G. (2002). Putting the community back into community-based resource management: A criteria and indicators approach to sustainability. *Human Organization* 61: 350–363.
- Pinkerton, E. (1989). Introduction: Attaining better fisheries management through co-management—Prospects, problems, and propositions. In Pinkerton, E. (ed.), *Co-operative Management of Local Fisheries: New Directions for Improved Management and Community Development*, University of British Columbia Press, Vancouver, Canada, pp. 3–33.
- Pinto, E. F. (2000). *Contesting Frontier Lands in Palawan, Philippines: Strategies of Indigenous Peoples for Community Development and Ancestral Domain Management*, Unpublished

- M.A. Thesis, Department of International Development and Social Change, Clark University, Worcester, MA.
- Pomeroy, R. S. (1995). Community-based and co-management institutions for sustainable coastal fisheries management in Southeast Asia. *Ocean & Coastal Management* 27: 143–162.
- Pomeroy, R. S., and Pido, M. D. (1995). Initiatives towards fisheries co-management in the Philippines: The case of San Miguel Bay. *Marine Policy* 19: 213–226.
- Rivera, R., and Newkirk, G. F. (1997). Power from the people: A documentation of non-governmental organizations' experience in community-based coastal resource management in the Philippines. *Ocean and Coastal Management* 36: 73–95.
- Russell, S. D., and Alexander, R. T. (2000). Of beggars and thieves: Customary sharing of the catch and informal sanctions in a Philippine fishery. In Durrenberger, E. P., and King, T. D. (eds.), *State and Community in Fisheries Management: Power, Policy, and Practice*, Bergin & Garvey, Westport, CT, pp. 19–40.
- Seki, K. (2000). Social change and migration: A case from Palawan. In Ushijima, I., and Zayas, C. N. (eds.), *Bisayan Knowledge Movement and Identity*, Visayan Maritime Anthropological Studies III, Third World Studies Center, University of the Philippines, Quezon City, Philippines, pp. 271–294.
- Sunderlin, W. D., and Gorospe, M. L. G. (1997). Fishers' organizations and modes of co-management: The case of San Miguel Bay, Philippines. *Human Organization* 56: 333–343.
- Utting, P. (2000). An overview of the potential and pitfalls of participatory conservation. In Utting, P. (ed.), *Forest Policy and Politics in the Philippines*, Ateneo de Manila University Press, Quezon City, Philippines, pp. 171–215.
- White, A. T., Courtney, C. A., and Salamanca, A. (2002). Experience with marine protected areas planning and management in the Philippines. *Coastal Management* 30: 1–26.
- White, A. T., Hale, L., Renard, T., and Cortesi, L. (1994). *Collaborative and Community-Based Management of Coral Reefs: Lessons from Experience*, Kumarian Press, West Hartford, CT.
- Wingard, J. D. (2000). Community transferable quotas: Internalizing externalities and minimizing social impacts of fisheries management. *Human Organization* 59: 48–57.
- Yano, T. (1994). The characteristics of fisherfolk culture in Panay: From the viewpoint of fishing ground exploitation. In Ushijima, I., and Zayas, C. N. (eds.), *Fishers of the Visayas*, Visayan Maritime Anthropological Studies I: 1991–1993, College of Social Sciences and Philosophy, University of the Philippines, Quezon City, Philippines, pp. 3–51.
- Zerner, C. (2000). Introduction: Toward a broader vision of justice and nature conservation. In Zerner, C. (ed.), *People, Plants, and Justice: The Politics of Nature Conservation*, Columbia University Press, New York, pp. 3–20.